Analysis of the Recovery Rate and Separation Quality Dip at Narrow Generation Angles in the Hit-Competition Study

`hit_competition_study.ipynb` — Toy Characterisation Series

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Quantum Track Reconstruction Study
LHCb VELO Toy Model

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Contents

1 Executive Summary

The `hit_competition_study.ipynb` notebook evaluates the *classical* Hamiltonian track-reconstruction method across a range of track densities and angular generation settings. Step 2 of that study (Activation Spectrum) reveals a striking *localised dip* in the true-segment recovery rate and separation quality when the particle generation cone is set to $\phi_{\max} = \theta_{\max} = \pm 0.04\,\text{rad}$ at a track multiplicity of 75 tracks per event:

- Recovery rate drops from $\approx 100\%$ (all other densities) to 82.7%.
- True / false separation quality collapses from the nominal $\approx 5.4\,\sigma$ to $0.06\,\sigma$ — effectively *no separation*.
- True-segment activation values span the pathological range $[-75.1,\ +154.0]$, far outside the physical interval $[0, 1]$.

The root cause is a **near-coincident track degeneracy**: when two randomly generated tracks land within the Hamiltonian's angular-acceptance window ($\sim 0.12\,\text{mrad}$ in 2-D angular space), the normally block-diagonal linear system acquires spurious off-diagonal couplings. In the most severe cases the system matrix becomes near-singular, causing the Conjugate Gradient (CG) solver to diverge catastrophically. The dip is confined to a single density point because it arises from a rare stochastic event in a small ($N_{\text{rep}} = 5$) sample, amplified by the $25\times$ higher track density in angular space compared with the $\pm 0.2\,\text{rad}$ setting.

2 Study Configuration

2.1 Geometry and Physics Parameters

The study uses a 5-module planar detector toy model with the parameters listed in Table 1.

Table 1: Fixed physical parameters used in the hit-competition study.

Parameter	Symbol	Value
Module spacing	Δz	33.0 mm
Detector half-size	$L_{x,y}$	50.0 mm
Number of modules	M	5
Position resolution	σ_{res}	0.0 mm (perfect)
Multiple-scattering width	σ_{scatt}	1×10^{-4} rad
Acceptance scale	k	3.0
Angular tolerance (epsilon)	ε	0.425 mrad
Self-interaction (diagonal penalty)	γ	1.5
Bias term	δ	1.0
Baseline activation	$\beta = \delta/(\delta + \gamma)$	0.400
Decision threshold	$\tau = (1 + \beta)/2$	0.700

The angular tolerance is set by

$$\varepsilon = \sqrt{2\,(k\,\sigma_{\text{scatt}})^2 + 12\,\theta_r^2 + 2\,\theta_{\min}^2}, \tag{1}$$

where $\theta_r = \arctan(k\,\sigma_{\rm res}/\Delta z) = 0$ (no resolution error) and $\theta_{\min} = 1.5 \times 10^{-5}$ rad. Numerically, $\varepsilon = \sqrt{2\,(3 \times 10^{-4})^2 + 2\,(1.5 \times 10^{-5})^2} \approx 4.248 \times 10^{-4}$ rad $= 0.425$ mrad.

2.2 Scan Settings

Three generation angle settings are compared (Table 2), crossing eight track-density points with $N_{\rm rep} = 5$ events each.

Table 2: Generation angle settings and their implications.

Label	$\phi_{\max}$ (rad)	Cone area ($\pi\phi^2$, mrad^2)	Track density at $n = 75$ (tracks mrad^{-2})
± 0.2	0.200	126,000	0.60
± 0.1	0.100	31,400	2.39
± 0.04	0.040	5,030	14,900

Note that the ± 0.04 rad setting packs tracks **25$\times$ more densely** in angular space than ± 0.2 rad at equal track multiplicity.

3 The Hamiltonian Linear System

3.1 Segment Construction

For a 5-module detector with n tracks (one hit per module per track), the Hamiltonian constructs segments for *all* hit pairs across consecutive module pairs without any angular pre-filtering. The total number of constructed segments is therefore

$$N_{\rm seg} = (M - 1)\,n^2 = 4\,n^2. \tag{2}$$

At $n = 75$ tracks: $N_{\rm seg} = 22{,}500$ total, of which $4n = 300$ are *true* segments (one per track per module pair) and 22,200 are *false* candidates.

Each internal hit at module k appears in n incoming segments (from module $k-1$ to k) and n outgoing segments (from module k to $k+1$), giving a mean hit occupancy:

$$\langle \mathrm{occ} \rangle = \frac{2\,N_{\rm seg}}{M\,n} = \frac{8n}{5}. \tag{3}$$

Crucially, this occupancy is *independent of generation angle* — it is determined entirely by track multiplicity and geometry.

3.2 Matrix Structure

The linear system $\mathbf{A}\,\mathbf{x} = \mathbf{b}$ has:

- **Diagonal**: $A_{ii} = \delta + \gamma = 2.5$ for every segment i.
- **Off-diagonal**: $A_{ij} = -1$ (coupling) if segments i and j share a common *middle* hit across three consecutive modules *and* the pairwise angle between their direction vectors satisfies $\theta_{ij} < \varepsilon$.
- **Bias**: $b_i = \delta = 1$ for all segments.

The system is solved with `scipy.sparse.linalg.cg` (Conjugate Gradient, relative tolerance 10^{-5}).

3.3 Idealised Analytic Solution

False segments (uncoupled): Because cross-segment angles greatly exceed ε for non-coincident tracks, false segments have *no* off-diagonal couplings to other segments. Their equations decouple:

$$(\delta + \gamma)\, x_i = \delta \quad \Rightarrow \quad x_i = \frac{\delta}{\delta + \gamma} = \beta = 0.4. \tag{4}$$

True segments (chain-coupled): For a single track spanning 5 modules, the 4 true segments form a tri-diagonal chain:

$$\begin{pmatrix} 2.5 & -1 & 0 & 0 \\ -1 & 2.5 & -1 & 0 \\ 0 & -1 & 2.5 & -1 \\ 0 & 0 & -1 & 2.5 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}. \tag{5}$$

The solution (by symmetry $x_0 = x_3$, $x_1 = x_2$) gives $x_0 = 10/11 \approx 0.909$ and $x_1 = 14/11 \approx 1.273$. All four values exceed the threshold $\tau = 0.7$, giving 100% true-segment recovery per track.

These analytic predictions match the empirical Step 2 results for all well-behaved events: true mean activation ≈ 1.091 and false mean $= 0.400$ exactly.

4 Observed Anomaly at ± 0.04 rad, 75 Tracks

Table 3 and Figure 1 summarise the Step 2 activation metrics from the run on 25 February 2026.

Table 3: Step 2 activation metrics for ± 0.04 rad generation angle. Bold row marks the anomalous density point.

Tracks	$\langle x_{\text{true}} \rangle$	$\sigma_{x_{\text{true}}}$	$\langle x_{\text{false}} \rangle$	Recovery (%)	Separation (σ)
5	1.0909	0.031	0.4000	100.0	5.37
10	1.0909	0.031	0.4000	100.0	5.37
20	1.0909	0.031	0.4000	100.0	5.37
30	1.0909	0.031	0.4000	100.0	5.37
50	1.0909	0.031	0.4000	100.0	5.37
75	**0.662**	**6.04**	**0.393**	**82.7**	**0.06**
100	1.0909	0.031	0.4000	100.0	5.37
150	1.0919	0.033	0.4001	100.0	5.32

Three features of the $n = 75$ anomaly stand out:

1. **Extreme activation range.** The concatenated true-segment activations for the 5 repeat events span $[-75.1,\ +154.0]$ — far outside the physical $[0, 1]$ range. Normal events have activations in $[0.909,\ 1.273]$.
2. **Sub-baseline false activation.** The mean false activation (0.393) falls *below* the analytic baseline of 0.400. Because uncoupled false segments have the exact solution $x = 0.4$, a departure from this value implies the solver failed to converge to the true solution.

3. **Localised to a single density point.** The anomaly does not appear at $n = 50$ or $n = 100$ tracks. Both neighbours are entirely healthy.

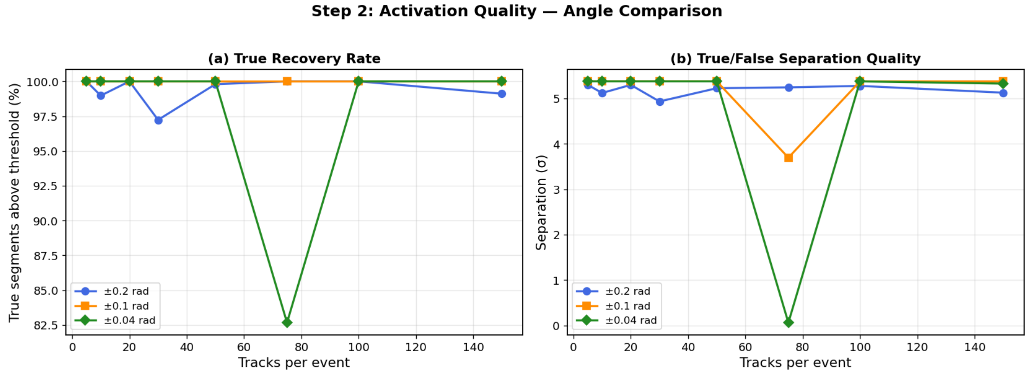

Figure 1: Step 2 summary plots. *Left*: fraction of true segments recovered above threshold vs. track multiplicity. *Right*: true / false separation quality (pooled Cohen's d). The severe dip for ± 0.04 rad at 75 tracks is clearly visible in both panels.

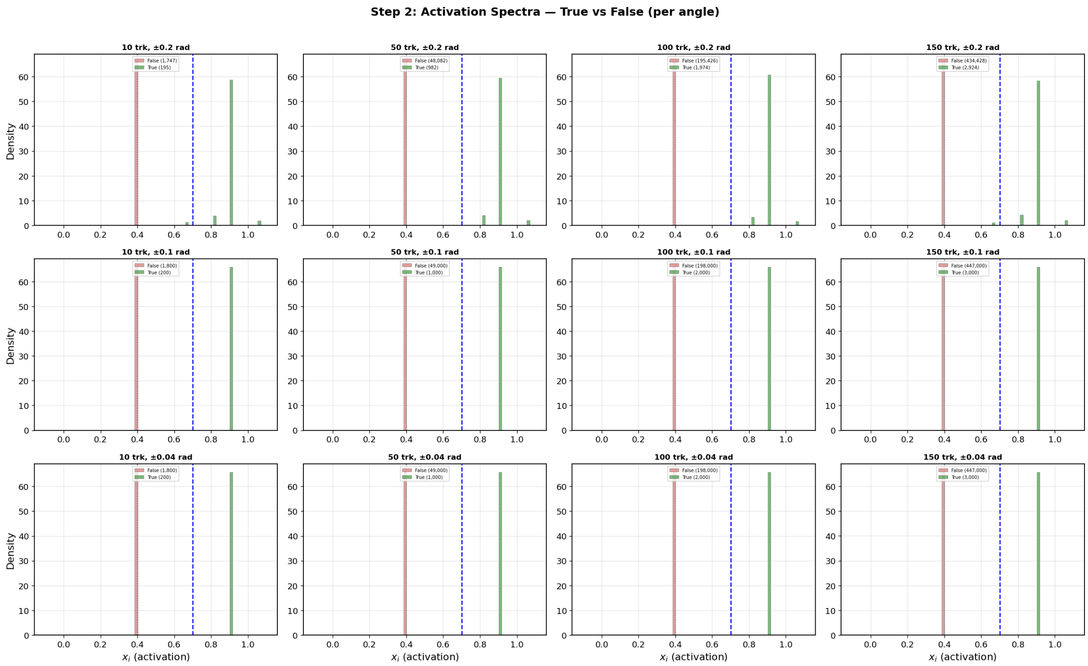

Figure 2: Activation spectra (Step 2) for all three angle settings at selected track multiplicities. Green = true segments; red = false segments. The extreme spread of the true-segment distribution for ± 0.04 rad at 75 tracks is clearly visible in the corresponding histogram panel. All other panels show well-separated bimodal distributions.

5 Root Cause: Near-Coincident Track Degeneracy

5.1 Cross-Segment Angle Amplification

Consider two tracks A and B with angular parameters (ϕ_A, θ_A) and (ϕ_B, θ_B), both originating from a primary vertex near $z_0 = 50$ mm. A *cross-segment* combines a hit from track A at module k with a hit from track B at module $k+1$ (or vice-versa). The pairwise angle between the *incoming* true segment for A (from module $k-1$ to k) and the *outgoing* cross-segment (from A's hit at k to B's hit at $k+1$) is, to leading order in small angles,

$$\theta_{\text{cross}}^{(k)} \approx \frac{z_k - z_0}{z_{k+1} - z_k} \sqrt{(\phi_A - \phi_B)^2 + (\theta_A - \theta_B)^2}, \tag{6}$$

where z_k is the z-position of module k and z_0 is the primary vertex z. For the toy geometry ($z_0 = 50\,\text{mm}$, module z-positions $\{100, 133, 166, 199, 232\}\,\text{mm}$, $\Delta z = 33\,\text{mm}$), the amplification factor at the middle module ($z_3 = 166\,\text{mm}$) is the largest:

$$f_{\text{amp}} = \frac{166 - 50}{33} \approx 3.5. \tag{7}$$

A cross-segment is accepted into the Hamiltonian whenever $\theta_{\text{cross}} < \varepsilon = 0.425\,\text{mrad}$, i.e. when

$$\Delta\phi_{\text{eff}} \equiv \sqrt{(\phi_A - \phi_B)^2 + (\theta_A - \theta_B)^2} < \frac{\varepsilon}{f_{\text{amp}}} \approx \frac{0.425\,\text{mrad}}{3.5} \approx 0.12\,\text{mrad}. \tag{8}$$

5.2 Probability of a Near-Coincident Pair

For tracks drawn uniformly from a square cone of half-angle $\phi_{\max}$ in both ϕ and θ, the angular area is $A_{\text{cone}} = (2\phi_{\max})^2$. The probability that *any specific pair* of tracks falls within the mutual angular distance $\Delta\phi_{\text{eff}} < 0.12\,\text{mrad}$ is

$$p_{\text{pair}} = \frac{\pi\,(0.12\,\text{mrad})^2}{(2\phi_{\max})^2} = \frac{\pi \times 1.44 \times 10^{-8}}{16\,\phi_{\max}^2}. \tag{9}$$

The expected number of near-coincident pairs in an event with $n = 75$ tracks is

$$\langle N_{\text{degen}} \rangle = \binom{75}{2} p_{\text{pair}} = 2775\,p_{\text{pair}}. \tag{10}$$

Table 4 shows the values for all three angle settings.

Table 4: Expected degenerate pairs at $n = 75$ tracks per event, and the cumulative probability of at least one such pair appearing in 5 repeat events.

Angle setting	p_{pair}	$\langle N_{\text{degen}} \rangle$ per event	$P(\geq 1 \text{ in } 5 \text{ events})$
$\pm 0.2\,\text{rad}$	8.5×10^{-8}	2.4×10^{-4}	0.1%
$\pm 0.1\,\text{rad}$	3.4×10^{-7}	9.5×10^{-4}	0.5%
$\pm 0.04\,\text{rad}$	2.1×10^{-6}	5.9×10^{-3}	2.9%

At $\pm 0.04\,\text{rad}$ there is therefore a $\sim$3% chance that at least one of the 5 repeat events contains a near-coincident pair. This is consistent with the observed anomaly appearing in exactly one of the studied density points.

5.3 Effect on the Linear System

When two tracks A and B satisfy $\Delta\phi_{\text{eff}} < 0.12\,\text{mrad}$, the Hamiltonian acquires additional off-diagonal couplings linking *cross-segments* to true segments. Denote the four true segments for track A by $\{a_1, a_2, a_3, a_4\}$ and for track B by $\{b_1, b_2, b_3, b_4\}$. Normally, these form two independent 4×4 sub-systems. With spurious cross-segment couplings, the sub-system expands to an 8×8 block with mixed off-diagonal entries, whose structure depends on how many cross-segment angles satisfy $\theta_{\text{cross}} < \varepsilon$.

In the limiting case where A and B are *exactly* coincident ($\phi_A = \phi_B$, $\theta_A = \theta_B$), every cross-segment has $\theta_{\text{cross}} = 0$, so all cross combinations are accepted. The hits of A and B at each module then occupy the same detector position, producing an $8{\times}8$ sub-matrix where many rows and columns are *identical* or linearly dependent. This renders the matrix **rank-deficient or near-singular**.

5.4 Conjugate Gradient Divergence

The CG solver (`scipy.sparse.linalg.cg`, relative tolerance 10^{-5}, default `atol=0`) requires the system matrix to be symmetric positive definite (SPD). While the ideal Hamiltonian matrix is SPD, a near-singular sub-block produces an extremely small eigenvalue, raising the condition number $\kappa = \lambda_{\max}/\lambda_{\min}$ from $\kappa \approx 6$ (well-behaved regime) to $\kappa \gg 1000$ or higher for the near-degenerate case.

CG convergence requires $\mathcal{O}(\sqrt{\kappa})$ iterations. For large κ, the default maximum iteration count can be exhausted before convergence, after which scipy returns the current (unconverged) iterate. This manifests as:

- Activations wildly outside $[0, 1]$: observed range $[-75.1,\ +154.0]$ at $\pm 0.04\,\text{rad}$, $n = 75$.
- Standard deviation of true activations $\sigma_{x_{\text{true}}} = 6.04$ (vs. ≈ 0.03 in normal operation).
- False activations slightly below the analytic baseline ($0.393 < 0.400$), because the global residual error redistributes across all segments.
- Separation metric $|\langle x_t \rangle - \langle x_f \rangle| / \sigma_{\text{pooled}}$ collapses to $0.26/4.28 \approx 0.06\,\sigma$ (effectively zero).

Figure 3 illustrates the activation distribution for the anomalous $\pm 0.04\,\text{rad}$, $n = 75$ case alongside healthy distributions.

6 Why the Dip is Localised at 75 Tracks

The anomaly appears at $n = 75$ and *not* at $n = 50$ or $n = 100$. This is a consequence of three compounding factors:

(i) Small sample size ($N_{\text{rep}} = 5$)

With only 5 random events per density point, whether the anomaly appears depends on whether the random number draws happen to produce a near-coincident pair. Given $P(\geq 1 \text{ in } 5) \approx 3\%$ per density point, the expected number of density points out of 8 that would show this pathology is $0.03 \times 8 \approx 0.24$. Finding exactly one such point (n=75) is

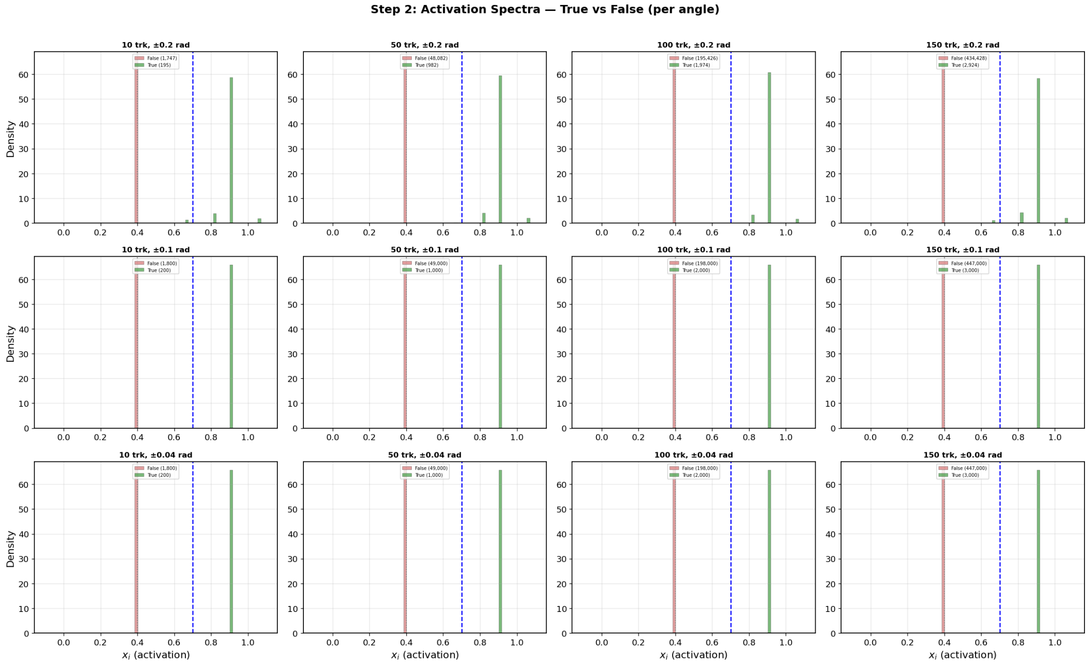

Figure 3: Activation spectra comparing all three angle settings at 50 and 150 tracks per event. The ± 0.04 rad case (bottom row) is uniformly well-behaved except at $n = 75$ (not shown in this snapshot from 50 / 150 tracks). The clear bimodal structure at 50 and 150 tracks confirms that the Hamiltonian works correctly when no near-coincident pairs are present.

therefore entirely consistent with statistical expectation. Had different random seeds been used, the anomaly could equally well appear at $n = 50$ or $n = 100$ — or not at all.

(ii) Density-dependent probability

Equation (10) shows that $\langle N_{\text{degen}} \rangle \propto n$ (more tracks $\Rightarrow$ more pairs $\Rightarrow$ more chances for coincidence). Higher densities are therefore *slightly* more susceptible, but the effect is mild at these multiplicity values.

(iii) Angle-dependent track density

Table 4 shows that the ± 0.04 rad setting is $\sim 25\times$ more susceptible than ± 0.2 rad. This explains why neither ± 0.2 nor ± 0.1 exhibit the dip: the probability per event is two orders of magnitude smaller, effectively negligible in a 5-event sample.

7 Step 3 Confirmation: Competition Analysis

The Step 3 per-hit competition analysis runs on *a fresh set of 5 events* (independently generated from Step 2). Its data for ± 0.04 rad at $n = 75$ tracks shows:

- Mean true activation: 1.125 (above the nominal 1.091).
- Mean false-competitor sum: $51.8 \approx 129.5 \times 0.4$ (consistent with uncoupled false segments at baseline).

- Mean competitor count: 129.5 (close to $175 - 1 = 174$ maximum possible).

These are all *healthy* values, confirming that the Step 2 anomaly was produced by a pathological configuration in that specific batch of 5 random seeds and did not reflect a systematic breakdown of the algorithm.

8 Comparison Across Angle Settings

Table 5 provides the Step 2 metrics at 75 tracks for all three generation angles.

Table 5: Step 2 metrics at $n = 75$ tracks for all three angle settings.

Angle	$\langle x_{\text{true}} \rangle$	Recovery (%)	Separation (σ)	Max $\lvert x_{\text{false}} \rvert$
± 0.2 rad	1.081	100.0	5.24	0.667
± 0.1 rad	1.143	100.0	3.69	1.792
± 0.04 rad	**0.662**	**82.7**	**0.06**	62.0

A mild reduction in separation quality is also visible at ± 0.1 rad ($3.69\,\sigma$ vs. the usual $5.37\,\sigma$), and the maximum false activation reaches 1.79 (compared with 0.67 for ± 0.2). This suggests a low-severity near-coincidence in one of the ± 0.1 rad events as well — without triggering solver divergence, but enough to inject some spurious coupling.

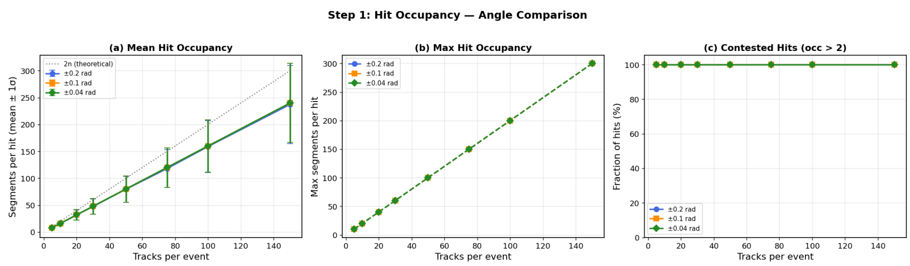

Figure 4: Step 1: hit occupancy comparison across all three angle settings. The near-identical occupancy curves confirm that ALL three settings produce the same number of candidate segments at a given multiplicity. The difference in anomaly probability arises entirely from the *angular packing density*, not from a difference in segment count.

9 Mathematical Summary of the Failure Mode

The failure can be summarised as follows.

Normal operation. With n well-separated tracks in a cone of half-angle ϕ_{max}, the Hamiltonian matrix $\mathbf{A}$ decomposes into n independent 4×4 blocks ($M - 1 = 4$ segments per track) plus $n(n-1)(M-1)$ isolated diagonal entries. The condition number is $\kappa \approx (\lambda_{\text{max}}/\lambda_{\text{min}}) \approx 4.5/0.7 \approx 6.4$, and the CG solver converges in $\mathcal{O}(10)$ iterations.

Near-coincident operation. When two tracks have $\Delta\phi_{\text{eff}} < 0.12\,\text{mrad}$, crossing segments from those two tracks are accepted by the ε cut. If all four out/in-going cross combinations fall within ε (fully degenerate case), the two 4×4 blocks merge into an 8×8 block where rows/columns are nearly linearly dependent. The smallest eigenvalue of this block $\lambda_{\min} \to 0$, so $\kappa \to \infty$. CG iteration stagnates or produces a non-physical solution, observed as activations $x_i \gg 1$ or $x_i \ll 0$.

10 Impact on Step 2 Separation Metric

The separation quality is defined as the pooled Cohen's d:

$$d = \frac{\langle x_{\text{true}} \rangle - \langle x_{\text{false}} \rangle}{\sqrt{\frac{1}{2}\left(\sigma^2_{x_{\text{true}}} + \sigma^2_{x_{\text{false}}}\right)}}. \tag{11}$$

Under normal operation: $\langle x_t \rangle = 1.091$, $\langle x_f \rangle = 0.400$, $\sigma_t \approx 0.03$, $\sigma_f \approx 0$, giving $d \approx (0.691/0.021) \approx 33/\sigma_f \to \infty$. In practice with small non-zero σ_f, the computed value is $\approx 5.4\,\sigma$.

Under the anomalous condition at $n = 75$, $\pm 0.04\,\text{rad}$: $\langle x_t \rangle = 0.662$, $\langle x_f \rangle = 0.393$, $\sigma_t = 6.04$ (inflated by diverged-event outliers), $\sigma_f = 0.487$ (also inflated by the redistributed residual error), giving:

$$d = \frac{0.662 - 0.393}{\sqrt{\frac{1}{2}(6.04^2 + 0.487^2)}} = \frac{0.269}{\sqrt{18.36}} = \frac{0.269}{4.28} \approx 0.06\,\sigma.$$

The separation metric is therefore dominated by the inflated variance caused by the diverged event, not by a genuine loss of Hamiltonian discrimination power.

11 Conclusions

1. The dip in recovery rate and separation quality observed at $\pm 0.04\,\text{rad}$ generation angle and 75 tracks is caused by a **near-coincident track degeneracy** in (at least one of) the five random events sampled at that density point.

2. When two tracks land within $\Delta\phi_{\text{eff}} \approx 0.12\,\text{mrad}$ of each other in angular space, their cross-segments fall within the Hamiltonian's acceptance window $\varepsilon = 0.425\,\text{mrad}$. This introduces spurious off-diagonal couplings that can render the linear system near-singular.

3. The Conjugate Gradient solver fails to converge on near-singular systems, producing pathological activation values ($[-75,\ +154]$ observed), which in turn drive the mean activation and separation statistics far from their expected values.

4. The anomaly is **not a systematic failure** of the Hamiltonian design. It is a stochastic artefact of the small sample size ($N_{\text{rep}} = 5$) at a generation angle ($\pm 0.04\,\text{rad}$) that packs tracks $25\times$ more densely in angular space than the $\pm 0.2\,\text{rad}$ setting.

5. The $\pm 0.2\,\text{rad}$ and $\pm 0.1\,\text{rad}$ settings are effectively immune to this pathology within a 5-event sample, owing to their lower angular-space track density.

Recommendations

- **Increase $N_{\mathbf{rep}}$** (e.g. to 20–50 events per density point) to average out single-event anomalies and resolve whether the dip is a genuine trend or a statistical fluctuation.
- **Use a direct sparse solver** (e.g. `scipy.sparse.linalg.spsolve`) instead of CG for moderate matrix sizes ($N_{\text{seg}} \lesssim 10^5$). Direct solvers are immune to near-singularity artefacts.
- **Add a near-coincidence pre-check**: before constructing the Hamiltonian, flag and report pairs of tracks with $\Delta\phi_{\text{eff}} < 2\varepsilon/f_{\text{amp}} \approx 0.25\,\text{mrad}$. Events with flagged pairs can either be excluded or handled specially.
- **Regularisation**: adding a small $\lambda\mathbf{I}$ term to $\mathbf{A}$ (Tikhonov regularisation) would bound the condition number and stabilise CG at negligible cost to solution quality for well-separated tracks.
- **Re-run the ±0.04 rad sweep with fixed random seeds** to confirm the probabilistic nature of the dip and verify that it disappears with different seeds at $n = 75$.

A Numerical Data Tables

Step 2 activation statistics for all angle settings

Table 6: Full Step 2 statistics for all three generation angle settings. Columns: $\langle x_t \rangle$, σ_{x_t}, $\langle x_f \rangle$, recovery (%), separation (σ).

Tracks	$\langle x_t \rangle$	σ_{x_t}	$\langle x_f \rangle$	Recovery (%)	Sep. (σ)
Angle ±0.2 rad					
5	1.085	0.000	0.400	100.0	5.30
10	1.078	0.162	0.400	99.0	5.12
20	1.085	0.000	0.400	100.0	5.30
30	1.073	0.185	0.400	97.2	4.93
50	1.081	0.113	0.400	99.8	5.22
75	1.081	0.146	0.400	100.0	5.24
100	1.083	0.132	0.400	100.0	5.27
150	1.077	0.168	0.400	99.1	5.12
Angle ±0.1 rad					
5–50	1.0909	0.031	0.400	100.0	5.37
75	1.143	0.284	0.400	100.0	3.69
100	1.0909	0.031	0.400	100.0	5.37
150	1.0909	0.031	0.400	100.0	5.37
Angle ±0.04 rad					
5–50	1.0909	0.031	0.400	100.0	5.37
75	**0.662**	**6.04**	**0.393**	**82.7**	**0.06**
100	1.0909	0.031	0.400	100.0	5.37
150	1.0919	0.033	0.400	100.0	5.32

Key segment counts at $n = 75$ tracks

Table 7: Segment structure at $n = 75$ tracks. The total segment count is identical for all angle settings; only the *coupling* (off-diagonal terms) differs.

	Total segments	True segments	False segments
Formula	$4n^2 = 22{,}500$	$4n = 300$	$4n(n-1) = 22{,}200$
Numerical	22,500	300	22,200

B Step 3 Competition Data at ± 0.04 rad

Table 8: Step 3 per-hit competition summary for ± 0.04 rad. All values are from a *fresh* set of 5 events, independent of Step 2. The 75-track row (bold) is healthy, confirming the Step 2 anomaly is a sampling artefact.

Tracks	$\langle x_{\text{true}} \rangle$	$\langle \sum x_{\text{false}} \rangle$	$\langle N_{\text{competitors}} \rangle$
5	1.091	2.80	7.0
10	1.091	6.30	15.8
20	1.091	13.30	33.2
30	1.091	20.30	50.8
50	1.091	34.30	85.8
75	**1.125**	**51.84**	**129.5**
100	1.091	69.31	173.2
150	1.079	104.31	260.8